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Frequency-band graph-based sensor fusion with sensitivity-aware energy assist network for machinery system fault diagnosis

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ABSTRACT

Graph-based fault diagnosis methods have shown capabilities in multi-sensor data fusion by representing the hidden relationships between sensors. However, their effective application in complex machinery systems remains challenging. First, constructing a graph that fully captures system-level fault characteristics is difficult, as fault-related features in signals are often subtle and widely distributed across multiple frequency bands. Second, learning accurate and efficient graph representations during model training is hindered by the fact that sensor signals exhibit varying sensitivities to different fault modes, and each fault mode presents distinct frequency-domain behaviors, such as energy shifts or unique fault frequencies. To overcome these challenges, this article proposes frequency-band graph-based sensor fusion with sensitivity-aware energy assist network (FBG-SEANet), a newly developed graph-based sensor fusion framework. The frequency-band graph (FBG) is a novel graph construction strategy that decomposes sensor signals into multiple frequency bands, builds a subgraph for each band, and connects them to form the final graph, thereby preserving subtle fault features distinctive to each band. Using the FBG as input, the sensitivity-aware energy assist network (SEANet) is designed with a sensitivity-aware readout (SR) layer to capture signal sensitivity to each fault. In addition, an energy assist (EA) module is employed to learn fault features associated with energy shifts in the signals. Validation is carried out using one open rotating machinery dataset, one motor experimental dataset, and one urban air mobility (UAM) system experimental dataset. The results show that FBG-SEANet improves multi-sensor data fusion and outperforms conventional sensor fusion models with enhanced explainability.

1. Introduction

Machinery systems are complex assemblies comprising multiple components that operate in unison, playing a crucial role in modern industries (Yang et al., 2023a; Li et al., 2022b; Miao et al., 2024). With ongoing technological advancements, such systems are increasingly being adopted in emerging fields such as urban air mobility (UAM) and electric cars (Straubinger et al., 2020; Sun et al., 2019; Dai et al., 2024). Failures of these machinery systems can result not only in significant economic losses but also in serious safety risks to users (Li et al., 2024; Tian et al., 2018; Wang et al., 2025a). Thus, developing reliable fault diagnosis methods is of great importance to ensure the stable operation of machinery systems (He et al., 2020; Feng et al., 2023b; Straubinger et al., 2020; Zhiyi et al., 2020).

Advances in signal acquisition technologies for fault diagnosis have enabled the collection of large volumes of monitoring data through various sensor modalities (Zhao et al., 2023). To effectively leverage such data, deep learning (DL)-based methods have emerged as a representative data-driven approach owing to their ability to autonomously extract and select relevant features. Representative examples include long short-term memory (LSTM) networks for motor fault diagnosis (Xiao et al., 2019), convolutional neural network (CNN)-based encoders for rolling bearing fault (Qin and Shi, 2022), variational autoencoders (VAEs) for signal dimensionality reduction in motor fault diagnosis

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(Huang et al., 2019), and transformer models for capturing global dependencies in rolling bearing fault diagnosis (Wang et al., 2024b).

Despite these advancements, applying DL-based methods to complex machinery systems remains challenging due to the intricate nature of such systems. These systems are composed of mechanical and electrical components that closely interact with each other (Choudhary et al., 2022), leading to strong interdependencies among signals from different subsystems. As a result, there is a growing need for system-level fault diagnosis that can account for these cross-component interactions. Conventional DL-based methods, however, typically process sensor data by simply concatenating signals and applying fixed convolutional kernels (Kim et al., 2023b). This limits their ability to model the non-linear and cross-sensor interactions that are intrinsic to complex machinery systems, thereby preventing them from fully leveraging the rich structural relationships embedded in the sensor data.

To address these challenges, many researchers have recently employed graph neural network (GNN)-based methods (Zhong et al., 2025a; Xiao et al., 2024) including a graph convolutional network (GCN) (Zhang et al., 2022a, 2024a; Gao et al., 2021) and a graph attention network (GAT) (Veličković et al., 2017; Chen et al., 2025) offering advantages in multi-sensor fusion for system-level fault diagnosis (Wang et al., 2024a; Feng et al., 2023a; Li et al., 2022a; Xu et al., 2024). The graph-based method can reveal hidden relationships between sensors by representing their interactions and features as edges and nodes (Zhao et al., 2024; Wei et al., 2024, 2025). This makes graph-based methods particularly effective for multi-sensor fusion, as each node aggregates information from neighboring sensors through message passing, enabling the model to jointly leverage the entire sensor network. Additionally, in the graph domain, the geometric structure of the graph can also represent the relationships between sensors, enabling better modeling of inter-sensor dependencies (Sun et al., 2021; Zheng et al., 2024).

Graph-based, system-level fault diagnosis frameworks can be broadly categorized into node-level and graph-level. The node-level approach represents an individual sample as each node in a graph (Li et al., 2022a; Zhang et al., 2024b). However, because such approaches mainly focus on local neighborhood information and their computational cost increases with the number of nodes, they are less suitable for large-scale graph data from multi-sensor machinery systems (Wang et al., 2024a). In contrast, graph-level approaches represent an entire system state as a single graph and learn a unified graph representation that captures structural relationships among acquisition signals (Wang et al., 2023; Zhong et al., 2025b). By integrating both local and global graph information, graph-level approaches are more effective for modeling complex inter-sensor correlations and are therefore more widely adopted in system-level fault diagnosis.

Leveraging these advantages, various graph-level methods have been proposed with different strategies for graph construction and representation. For graph construction, prior studies have built graphs from raw signals using similarity-based relations, from temporally processed features extracted by feature encoders, and from alternative signal representations such as Markov field graphs or frequency-domain features (Wang et al., 2023; Yang et al., 2023b; Han et al., 2024; Zhang et al., 2022b, 2023). For graph representation, attention mechanisms have been incorporated during graph convolution to capture the relative importance of different sensors, followed by a readout layer to obtain the final graph-level representation (Zhang et al., 2023; Wang et al., 2023; Yang et al., 2023b).

Despite the progress of these studies, several issues still remain in the context of system-level fault diagnosis, particularly in both graph construction and graph representation. First, in terms of graph construction, directly using raw sensor signals for node embedding presents some challenges. Dominant operational components in raw signal can mask subtle fault features, and since raw time-series signals contain overlapped frequency information, it is difficult to capture fault characteristics distributed across multiple frequency bands. Furthermore, methods utilizing energy spectrum from the raw signal typically aggregate the entire spectrum at once into a single vector, which can still dilute fault features present in narrow frequency bands.

Second, in terms of graph representation, representing distinct fault features presents some challenges. Applying attention mechanisms only in the convolution stage, rather than the readout stage, may not fully incorporate inter-sensor importance, as the readout layer is the final stage for graph representation. This may lead the model to suffer from information redundancy, thereby failing to effectively represent fault features. Moreover, in practical systems, each fault mode exhibits distinct fault behaviors, such as energy variations in pre-existing frequencies or the appearance of unique fault frequencies. However, existing methods did not fully consider these unique fault behaviors, limiting their robustness in diagnosing various fault modes.

To address these issues, this article proposes a frequency band graph-based sensor fusion with sensitivity-aware energy assist network (FBG-SEANet), a novel graph-based, system-level fault diagnosis framework that integrates a new graph construction strategy with an enhanced graph representation network. The proposed method begins with two-stage frequency-band graph (FBG) construction based on signal processing. In the first stage, a notch filter is applied in operation component filtering (OCF) to remove operational components from the signals. In the second stage, band-pass filters are utilized to generate subgraphs for each frequency band, thereby ensuring that the graph reflects more detailed and band-specific fault characteristics in the frequency domain. To effectively represent the constructed graph, FBG-SEANet further incorporates two specialized modules. First, a sensitivity-aware readout layer (SR layer) is introduced to fully incorporate inter-sensor importance, reducing information redundancy and enhancing interpretability. Second, an energy assist (EA) module embeds energy-related features into each node, allowing the model to adequately reflect energy variations in pre-existing frequency components caused by faults, thereby achieving reliable system-level fault diagnosis across various fault modes.

The main contributions are summarized as follows:

- Propose FBG-SEANet, a unified framework that jointly addresses graph construction and graph representation for system-level fault diagnosis.
- Develop the FBG, a novel graph construction strategy that eliminate operational components and captures band-specific fault features.
- Design the SR layer and EA module to enhance representation learning by modeling sensor importance and incorporating energy shift information.

The remainder of the paper is organized as follows. Section 2 provides the theoretical background of graph-based fault diagnosis. Section 3 illustrates the proposed method, FBG-SEANet, in detail. Section 4 presents experimental validation and discussions based on three datasets. Finally, Section 5 summarizes the conclusions of the paper.

2. Theoretical background

2.1. Graph convolutional networks

GCNs have emerged as an effective tool for learning representations from graph-structured data, characterized by nodes as entities and edges as the connections between them (Zhang et al., 2019; Shen et al., 2026). The key concept behind GCNs is the aggregation of features from neighbor nodes to update each node representation (Zhao et al., 2022; Shen et al., 2026). The graph convolution operation is a key component of this process, and it can be formulated in various ways depending on the specific model and approximation techniques. The graph convolution operation is generally expressed as the application of a filter g to a graph signal $x \in \mathbb{R}^n$. Here n represents the number of nodes, and mathematically defined as:

$$\mathbf{h}_{\text{graph convolution}} = \mathbf{G}(x, g) = \mathbf{U}\left(\mathbf{U}^{\mathrm{T}}x \odot \mathbf{U}^{\mathrm{T}}g\right) \tag{1}$$

where **G** represents the graph convolution, and **U** is the eigenvector matrix of the symmetric normalized graph Laplacian defined as $\mathscr{L} = \mathbf{I}_n - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$, where **D** and **A** refer to the degree matrix and adjacency matrix of the graph, respectively. The operator $\odot$ denotes the element wise Hadamard product. The matrix **U** contains the eigenvectors of the graph, while the filter g is applied to the graph signal x in the spectral domain. However, applying the graph convolution as defined above can be computationally expensive, especially for large graphs. To reduce the computational complexity, Defferrard et al. (2016) proposed the use of Chebyshev polynomials to approximate the graph filter, leading to the development of the ChebyNet model. The graph convolution in ChebyNet is expressed as:

$$g_\theta = \sum_{k=0}^{K-1} \theta_k T_k(\overline{\mathscr{L}}) \tag{2}$$

where $\overline{\mathscr{L}} = \frac{2\mathscr{L}}{\lambda_{max}} - \mathbf{I}_n$ is the rescaled graph Laplacian, and $T_k(\overline{\mathscr{L}})$ are the Chebyshev polynomials of order k. The Chebyshev polynomials are recursively defined as:

$$T_0(\overline{\mathscr{L}}) = \mathbf{I}_n,\ T_1(\overline{\mathscr{L}}) = \overline{\mathscr{L}},\ T_k(\overline{\mathscr{L}}) = 2\overline{\mathscr{L}}T_{k-1}(\overline{\mathscr{L}}) - T_{k-2}(\overline{\mathscr{L}}) \tag{3}$$

This approximation simplifies the graph convolution operation, as it reduces the need for explicit eigenvector computations while still capturing the essential properties of the graph. The graph convolution in ChebyNet can then be written as:

$$\mathbf{h}_{\text{ChebyNet}} = \mathbf{U}g_\theta\mathbf{U}^{\mathrm{T}}x = \sum_{k=0}^{K-1} \theta_k T_k(\overline{\mathscr{L}})x \tag{4}$$

This formulation makes the graph convolution process more computationally efficient, as it relies on a finite sum of Chebyshev polynomials rather than the full eigenvalue decomposition. Further simplifying the graph convolution, Kipf and Welling (2016) introduced the GCN, which approximates the graph convolution even more. GCN simplifies this further by using a normalized adjacency matrix, which significantly reduces the complexity and makes training scalable to large graphs. By assuming that the largest eigenvalue of the normalized graph Laplacian $\lambda_{max} \approx 2$, the filter in GCN simplifies to:

$$\mathbf{h}_{\text{simplified graph convolution}} = \theta_0 x + \theta_1(\mathscr{L} - \mathbf{I}_n)x = \theta_0 x + \theta_1\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}x \tag{5}$$

For multi-layer GCNs, the feature representations are updated layer by layer. The hidden layer activations are computed as:

$$\mathbf{H} = \sigma\left(\widetilde{\mathbf{D}}^{-1/2}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-1/2}\mathbf{X}\boldsymbol{\Theta}\right) \tag{6}$$

where $\widetilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}_n$ is the adjacency matrix with added self-loops, and $\widetilde{\mathbf{D}}$ is the corresponding degree matrix with $\widetilde{\mathbf{D}}_{ii} = \sum_j \widetilde{\mathbf{A}}_{ij}$. **X** represents the matrix of input feature, **H** is the matrix of output node features, $\boldsymbol{\Theta}$ is the matrix of learnable parameters, and σ represents a non-linear activation function.

2.2. Graph-level fault diagnosis

In graph-level fault diagnosis the entire system is modeled as a graph, enabling a fault diagnosis not only at the component level but also at the system level. Unlike node-level approaches that represent faults as simple nodes (Sun et al., 2024; Gao et al., 2021), this method considers the complex interactions between all nodes and edges, making it particularly suitable for system-level fault diagnosis. To construct system graphs, K-nearest neighbor graph (KNNGraph), and RadiusGraph are widely used (Li et al., 2022a; Zhang et al., 2025). With graphs constructed through these methods, the entire graph is treated as a single sample, and the training dataset consists of a collection of graphs. During the training process, the model learns to represent each entire graph as a single representation, capturing the changes in graph structure and information caused by system faults.

$$\mathbf{h}_{\text{Graph-level representation}} = \text{READOUT}(\{\mathrm{h}_v | v \in \mathrm{V}\}) \tag{7}$$

where h_v is the feature vector in **H**, V is the number of nodes and READOUT denotes function that aggregates the node representation.

This process involves the model learning both node and edge information across the entire graph. During training, the graph is reduced through pooling layers, and a readout layer compresses the pooled graph into a single representation. Therefore, changes in the graph structure due to fault are reflected in the reduced representation. Fig. 1 demonstrates how a graph that effectively captures fault characteristics changes when a system fault occurs, while Fig. 2 provides a simplified illustration of the training process for graph-level fault diagnosis.

To ensure the effectiveness of graph-level fault diagnosis, two critical factors should be considered. First, the graph should be constructed to accurately reflect the system's fault characteristics so that changes in nodes and edges due to the system fault can be effectively captured by the model. Second, since the entire graph is reduced to a single representation, it is essential to minimize information loss and redundancy. These considerations are key to achieving accurate and reliable fault diagnosis at the graph level.

2.3. System-level fault diagnosis

In system-level fault diagnosis, four characteristics should be mainly considered to make reliable and effective fault diagnosis models. First, as the fault components are subtle compared to operational components, significant changes in the acquired signal may not be evident (Kim et al., 2023a; Park et al., 2019, 2021). For example, in current signals, the supply frequency is a dominant operational component, and removing such operational components can improve fault sensitivity. Second, representative signals acquired from machinery systems, such as current, vibration, and acoustic signals, exhibit fault characteristics across different frequency bands (Smith and Randall, 2015; Jung et al., 2023). For example, for motor bearing fault with fault frequency at f_v in vibration signals appears at different frequency components in current signals. In current signal fault frequencies are modulated by the supply frequency f_s, having fault frequencies at $f_s \pm k\ f_v$ in current signals (Orlowska-Kowalska et al., 2022). Therefore, frequency-domain fault information should be considered to accurately capture system characteristics. Third, the contribution of each sensor varies depending on the fault type. In general, electrical faults tend to be more sensitive in current signals, whereas mechanical faults are more sensitive in vibration signals. For example, in motor systems, bearing and misalignment faults are more clearly reflected in vibration signals, while stator faults are more clearly reflected in current signals (Gangsar and Tiwari, 2020). This makes it important to distinguish and quantify sensor importance according to fault mode. Finally, different fault modes exhibit different fault behaviors. While some faults, such as bearing faults, generate clear fault-specific frequencies, others are primarily reflected as changes in the energy of pre-existing frequency components, such as the rotational frequency and its harmonics, that are already present under normal operating conditions. In such cases, the fault is manifested as a variation in the magnitude of these existing components rather than as the emergence of new fault-specific frequencies. For example, shaft misalignment and unbalance faults induce energy shifts in the rotational frequency and its harmonics (Patel and Darpe, 2009; Tahir et al., 2016). Such variation makes accurate diagnosis of diverse fault modes more challenging and highlights the need to account for fault-specific characteristics.

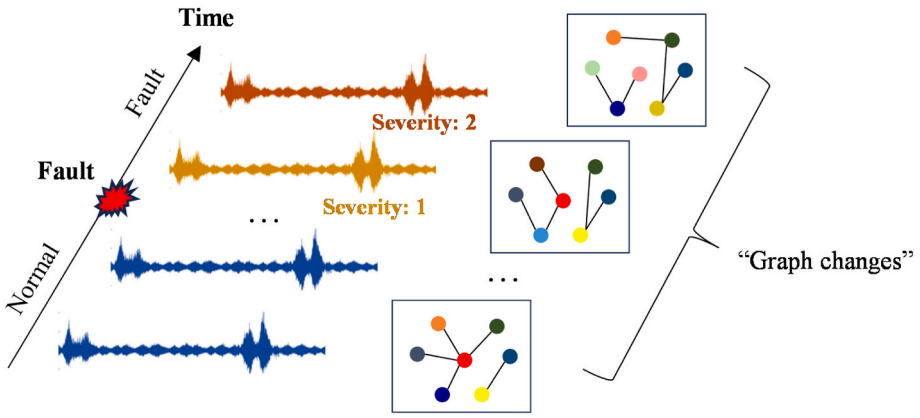


Fig. 1. An illustration of system graph changes due to fault.

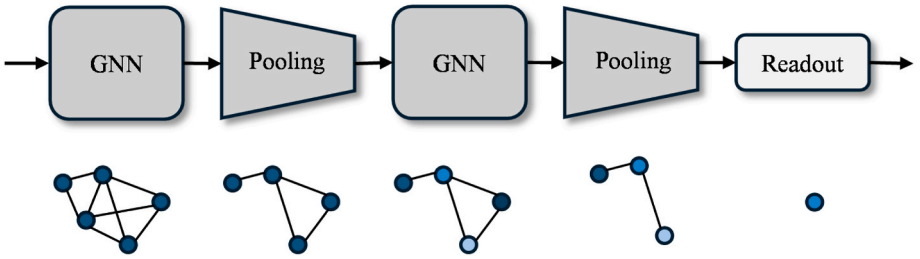


Fig. 2. An illustration of the graph-level graph representation process utilizing pooling and readout layer.

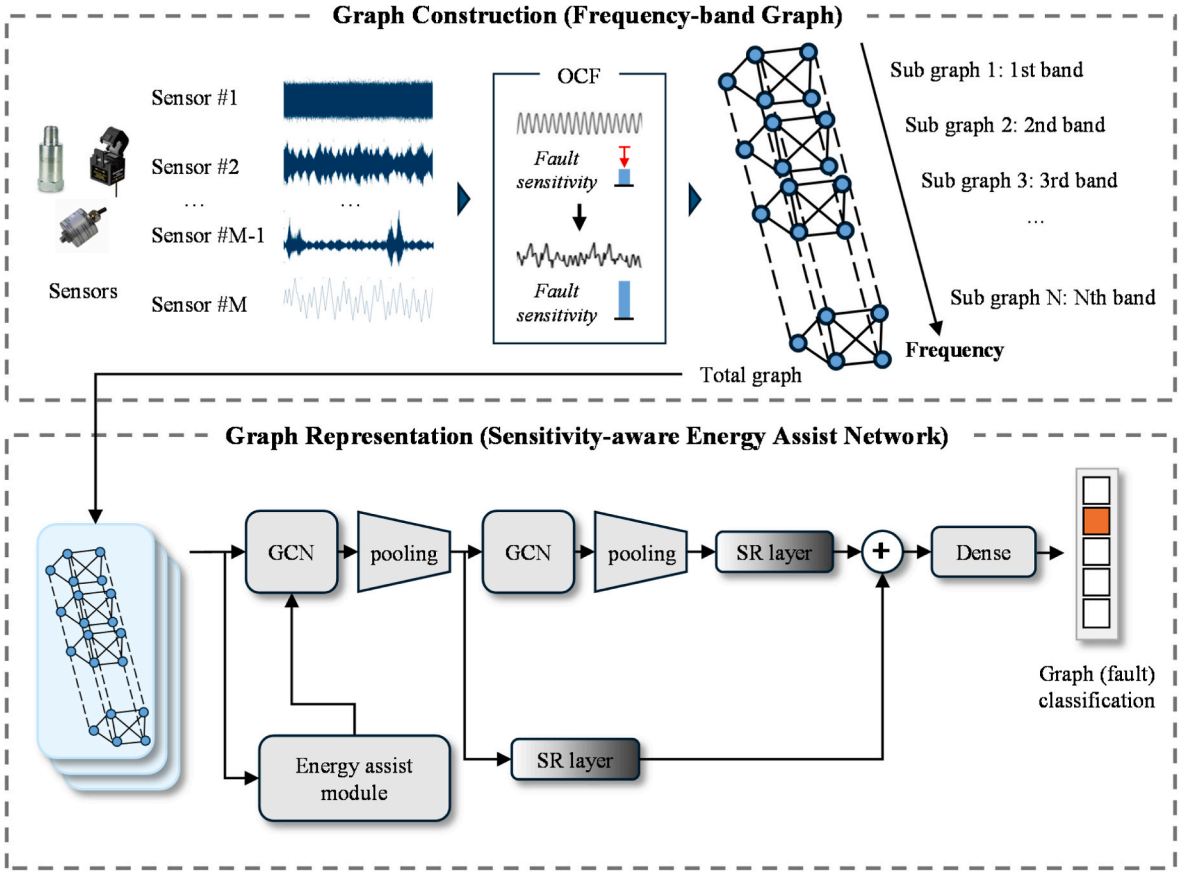


Fig. 3. Architecture of the proposed FBG-SEANet for graph-based, system-level fault diagnosis.

3. FBG-SEANet: frequency-band graph-based sensor fusion with sensitivity-aware energy assist network

3.1. Overall process

This study develops FBG-SEANet to tackle the following four key challenges in graph-based, system-level fault diagnosis:

- *Operational components mask fault features*: The FBG applies the OCF with a notch filter to remove dominant operational components from raw signals, enhancing the visibility of subtle fault-related features.
- *Fault information is spread across different frequency bands*: The FBG uses band-pass filtering to construct subgraphs for each frequency band and connects them into the final graph, preserving band-specific fault characteristics.
- *Sensor importance varies by fault mode*: The SR layer adaptively measures and weights sensor contributions, emphasizing the most relevant sensors for each fault type and reducing redundancy.
- *Energy shifts in pre-existing frequency components*: The EA module embeds energy-related features into each node, enabling the model to capture and utilize energy variations in pre-existing frequency components for accurate diagnosis.

The architecture of FBG-SEANet is shown in Fig. 3. It contains two main steps: (1) the graph construction step and (2) the graph representation step. In the graph construction step, multi-sensor signals are collected from the system and OCF is conducted to remove the effect of operational components. These preprocessed signals are then divided into frequency bands, and a subgraph is constructed for each frequency band, and the FBG is formed by connecting these subgraphs into the final graph. In the graph representation step, the EA module is employed to add energy features to nodes in input graphs, enriching the feature information. The input graphs are then trained using graph convolution layers, and the graph representation is pooled down using the SR layer, enabling a representation that considers the importance between the signals. Finally, fault classification is performed using the graph representation and a dense layer. The following subsections provide detailed descriptions of each component in the proposed FBG-SEANet.

3.2. Frequency-band graph construction

In graph-level fault diagnosis, constructing graphs that accurately represent the system's fault characteristics is of great importance. To achieve this, graph construction progresses through two primary stages: a signal preprocessing stage that removes operational components from the signals, and a graph construction stage that employs band-pass filters to create frequency band-specific graphs, which are then connected to form the final graph.

3.2.1. Operation component filtering

In the signal preprocessing stage, OCF is conducted to remove the effect of operational components, with a notch filter specifically tuned to the required frequencies. Fig. 4 provides a graphical illustration of the notch filter used in the OCF process. In this figure, f_{null} denotes the average cutoff frequency, which is aligned with the system's operational frequency. This allows OCF to reduce large operating components within the signal, thereby emphasizing the relatively small fault characteristics. This process helps to reflect the system's fault characteristics during the graph construction.

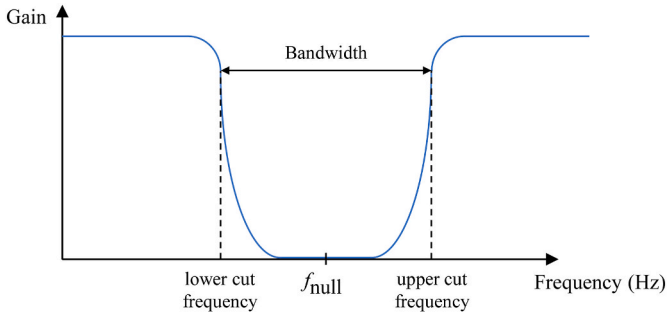


Fig. 4. Notch filter illustration.

3.2.2. Frequency-band graph

Signals acquired from machinery systems exhibit fault characteristics distributed across different frequency bands. However, these characteristics are often not clearly expressed in raw signals due to the overlapping of information from multiple frequency bands. To address this issue, the proposed FBG is constructed by decomposing the acquired signal into multiple frequency bands and creating a subgraph for each band. The overall process of FBG construction is shown in Fig. 5.

The construction of the FBG relies on determining two critical parameters: the bandwidth (B_{width}) and the number of frequency bands (N). These parameters are determined based on a generic kinematic guideline designed to balance signal processing stability and computational efficiency. The governing rule for selecting the bandwidth involves a discrete step selection with safety margin. To ensure practical applicability and standardized resolution, the bandwidth is selected in 10 Hz increments (e.g., 10, 20, 30 Hz). The specific value is chosen as the largest multiple of 10 that satisfies a safety margin constraint relative to the fundamental rotational frequency (f_r) of the target system:

$$B_{width} \leq \alpha \cdot f_r \tag{8}$$

Here, α is a safety coefficient established as 0.7. This constraint ($B_{width} \approx 0.7 \cdot f_r$) ensures that the fundamental frequency (1×) is positioned securely within a band, avoiding the edge distortions and spectral leakage that can occur if B_{width} is too close to f_r (e.g., if $B_{width} \approx 0.9 \cdot f_r$). Moreover, this constraint guarantees that the fundamental component (1×) and its subsequent harmonics (2×, 3×, etc.) are allocated to distinct frequency bands. This separation prevents multiple harmonic components from merging into a single node, thereby enabling the model to explicitly learn the inter-harmonic dependencies which are vital for diagnosing faults. Simultaneously, selecting the largest valid multiple of 10 maximizes the bandwidth size, thereby minimizing the total number of graph nodes and optimizing the computational cost for model training. For instance, for a system operating at 3000 rpm ($f_r \approx 50$ Hz), 30 Hz was selected as the bandwidth. This setting effectively separates the harmonics while maintaining a compact graph structure. This bandwidth design provides a systematic frequency-band selection rule based on the rotational characteristics of the target system rather than an empirical hyperparameter choice. For a given stationary operating condition, the bandwidth and number of frequency bands are systematically determined from the corresponding rotational frequency. Thus, the FBG construction is not tied to a single operating speed, but can be adapted to different stationary operating conditions by recalculating the bandwidth parameters according to the target rotational frequency.

Once the bandwidth is determined, the number of frequency bands (N) is derived based on a frequency threshold ($F_{threshold}$). This threshold is set to 1000 Hz to distinguish between the low-frequency range up to 1000 Hz, which requires fine-grained nodes to capture deterministic fault features (e.g., harmonics and sidebands), and the high-frequency range above 1000 Hz, which is dominated by broadband stochastic features suitable for aggregation. Accordingly, the signal is divided into N-1 bands of width B_{width} up to $F_{threshold}$, and the final N-th band captures all remaining high-frequency components.

Utilizing the determined bandwidth (B_{width}) and the number of frequency bands (N), the detailed graph construction process is as follows:

First, for each of the M channels of multi-modal signals preprocessed by the OCF $\{X_1, X_2, \ldots X_M\}$, band-pass filters are applied to divide each

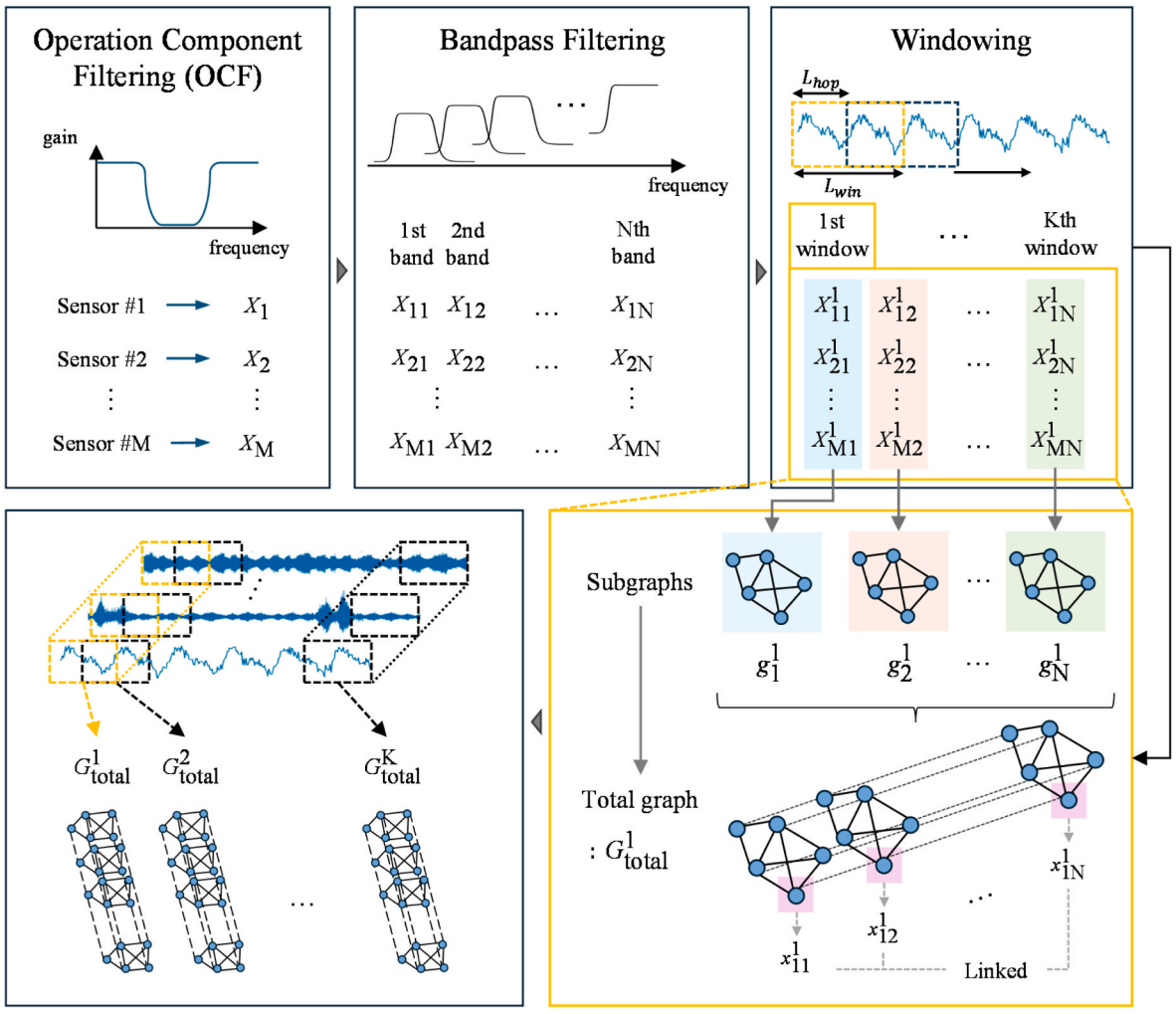


Fig. 5. The overall process of FBG construction.

signal into N frequency bands, resulting in a total of M × N signals $\{X_{11}, X_{12}, \ldots X_{1N}\}$, $\{X_{21}, X_{22}, \ldots X_{2N}\} \ldots \{X_{M1}, X_{M2}, \ldots X_{MN}\}$, where M is the number of sensors. Specifically, band-pass filters are applied to generate the first N-1 bands, whereas a high-pass filter is employed for the N-th band to retain all remaining high-frequency components. For example, if the bandwidth is set to 30 Hz and the number of bands N is 34, the first band ranges from 0 to 30 Hz, the second band from 30 to 60 Hz, and the final 34th band includes all signals above 990 Hz. Next, a sliding window is applied to segment each of these filtered signals into fixed-length signal samples x^k_{mn} ($m = 1, 2, 3, \ldots, M$; $n = 1, 2, 3, \ldots, N$; $k = 1, 2, 3, \ldots, K$). K is the number of segmented samples. For each segment, a graph is constructed, resulting in a total of K final graphs G^k_{total} ($k = 1, 2, 3, \ldots, K$). For example, to construct G^1_{total}, the first segments x^1_{mn} ($m = 1, 2, 3, \ldots, M$; $n = 1, 2, 3, \ldots, N$) are used. Specifically, using the first frequency band of the M channels $\{x^1_{11}, x^1_{21}, \ldots x^1_{M1}\}$, the first subgraph g^1_1 is constructed using the KNNGraph method (Li et al., 2022a). Similarly, using the second frequency band $\{x^1_{12}, x^1_{22}, \ldots x^1_{M2}\}$, the second subgraph g^1_2 is constructed, and this process is repeated until the N-th subgraph g^1_N is constructed using $\{x^1_{1N}, x^1_{2N}, \ldots x^1_{MN}\}$. Finally, the N band-specific subgraphs $\{g^1_1, g^1_2, \ldots, g^1_N\}$ are connected to form the final graph G^1_{total} for the first segmentation. When connecting these subgraphs, nodes within the same acquisition signal's frequency bands, such as $\{x^1_{11}, x^1_{12}, \ldots x^1_{1N}\}$, are connected, and similarly for other channels. This is because, although the signals are divided by frequency bands, they originate from a single acquisition signal X_1, and therefore, nodes representing different frequency bands of the same acquisition signal can be connected by edges. The resulting K final graphs G^k_{total} ($k = 1, 2, \ldots, K$) are then utilized for model training and testing.

The pseudocode of the proposed FBG construction is summarized below:

Algorithm 1. FBG construction process

Input: Multi-modal sensor signals preprocessed by the OCF (M sensors) $\mathbf{X} = \{X_1, X_2, \ldots X_M\}$, Number of frequency bands N, Bandwidth size B_{width}, Sliding window length W_{len}
Output: The set of frequency-band graphs $G_{set} = \{G^1_{\text{total}}, G^2_{\text{total}}, \ldots G^K_{\text{total}}\}$
//Stage 1: Signal Decomposition
for each sensor $m \leftarrow 1$ to M **do**
  for each frequency band $n \leftarrow 1$ to N **do**
    if $n < N$ then
      $X_{mn} \leftarrow$ BandPassFilter (X_m, cutoff frequency: $(n\text{-}1)B_{width}$, $(n)B_{width}$)
    else
      $X_{mn} \leftarrow$ HighPassFilter (X_m, cutoff frequency: $(N\text{-}1)B_{width}$)
    end if
  end for
end for
//Stage 2: Segmentation and Graph Construction
Segment all $\{X_{mn}\}$ into K samples $\{x^k_{mn}\}$ using window W_{len}
for each sample segment $k \leftarrow 1$ to K do
  Initialize node set V^k, edge set E^k, and subgraph set $S^k_{sub} \leftarrow \varnothing$
//Step 2-1: Construct Band-specific Subgraphs
  for each frequency band $n \leftarrow 1$ to N do
    Define node features $U^k_n = \{x^k_{1n}, x^k_{2n}, \ldots x^k_{Mn}\}$

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    Generate intra-band edges E_n^k using KNNGraph on U_n^k
    Construct subgraph g_n^k = (U_n^k, E_n^k)
    S_stub^k ← S_stub^k ∪ {g_n^k}
    Update global sets: V^k ← V^k ∪ U_n^k, E^k ← E^k ∪ E_n^k
  end for
 //Step 2-2: Connect Subgraphs (Inter-band)
  for each sensor m ← 1 to M do
    for each frequency band n ← 1 to N-1 do
      Select node u ∈ g_n^k (feature x_mn^k) and node v ∈ g_{n+1}^k (feature x_{m(n+1)}^k)
      Add inter-band edge (u, v) to E^k
    end for
  end for
  Form final graph G_total^k = (V^k, E^k) and add to G_set
end for
return G_set
```

3.3. Sensitivity-aware readout layer

The SR layer is constructed in the proposed SEANet using the self-attention method to better reflect the importance among signals during the graph representation extraction process in graph-level fault diagnosis. Fig. 6 shows the architecture of the SR layer. The attention mechanism is implemented to reflect the differences in contributions among the sensors which contribute differently depending on the type of fault. For implementation, in the SR layer, attention scores are assigned to nodes and spatial attention is imposed. Following a standard multi-head attention formulation, the SR layer first computes unnormalized attention scores from the query and key representations (Wang et al., 2025b). The scoring function for the attention mechanism is as follows:

$$\text{scoring function} = \frac{QK^{\mathrm{T}}}{\sqrt{d_k}} \tag{9}$$

where Q and K refer to the Query and Key vector in the attention mechanism, respectively, and $Q \in \mathrm{R}^{\mathrm{n}\times d_k}$, $K \in \mathrm{R}^{\mathrm{n}\times d_k}$ where d_k refers to the dimension of the K vector. Then, the attention weights can be obtained by feeding the score (z_i) into the Softmax function.

$$\mathrm{Softmax}(z_i) = \frac{\exp(z_i)}{\sum_{j=1}^{n}\exp(z_i)} \tag{10}$$

In the multi-head attention mechanism, the attention computation is independently performed for each head using its own query, key, and value representations, and the resulting head-wise attention outputs are then fused to obtain the final attention weight (Wang et al., 2025b). In the proposed SR layer, the final attention weights are derived by averaging the weights calculated by each head, where h denotes the number of attention heads in the multi-head attention mechanism.

$$\text{Multi-head attention weight} = \frac{1}{h}\sum_{i=1}^{\mathrm{h}}\text{attention weight of head}_i \tag{11}$$

Based on the final attention weights, weighted node representations are aggregated in the readout layer to form the final graph representation, thereby enabling graph representation considering signal importance.

3.4. Energy assist module

In order to diagnose some faults that have energy shift as a fault characteristic, the EA module is implemented in the proposed SEANet. These faults are not characterized by unique fault frequencies but rather by energy variations in pre-existing frequency bands, making them more difficult to diagnose. To effectively capture such energy variation, the root mean square (RMS) is utilized, as it is more effective than other statistical measures in representing the overall energy level of a signal.

Given an input graph, the EA module computes the RMS value from the time-domain signal of each node and embeds the energy information into the graph node features. Specifically, for node i with band-pass filtered signal x_{i} of length L, the RMS value is computed as:

$$\mathrm{RMS}_i = \sqrt{\frac{1}{\mathrm{L}}\sum_{j=1}^{\mathrm{L}} x_{i,j}^2} \tag{12}$$

The scalar RMS value is then projected into a d-dimensional (d = 16) energy feature vector through a dense layer:

$$\mathrm{e}_i = \mathrm{ReLU}(w\mathrm{RMS}_i + b) \tag{13}$$

where w and b are learnable parameters. Finally, the energy feature is added to the original node feature to produce the EA-enriched node representation:

$$\mathrm{h}_i' = \mathrm{h}_i + \mathrm{e}_i \tag{14}$$

where h_i is the original node feature obtained from the FBG construction and h_i' is the updated representation used as input to the subsequent GCN layers. In this way, the EA module bridges the time-series and graph domains by embedding signal energy information into the graph node features prior to graph convolution. Fig. 7 shows the illustrations of the EA module. Fig. 7 (a) shows the architecture of the EA module. Given an input graph, the EA module computes the RMS value of each node's signal, and these values are then passed through a dense layer to extract energy features for each node. Fig. 7 (b) compares fault-frequency-band signals under normal and unbalance fault conditions. Each signal is obtained by band-pass filtering to 90–120 Hz. Since an unbalance fault induces an energy shift in the pre-existing rotational-frequency sideband near 110 Hz, the fault signal exhibits higher amplitude within this band. These energy variations are extracted as features by the EA module and embedded into each node enabling the graph to effectively represent fault characteristics reflected in energy shift of pre-existing frequency components.

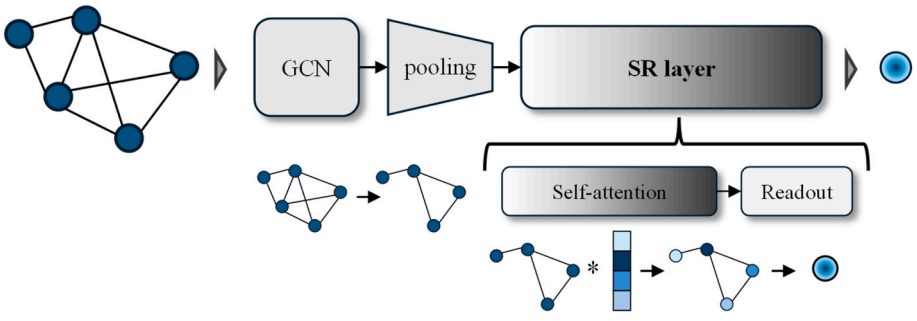


Fig. 6. Architecture of the SR layer.

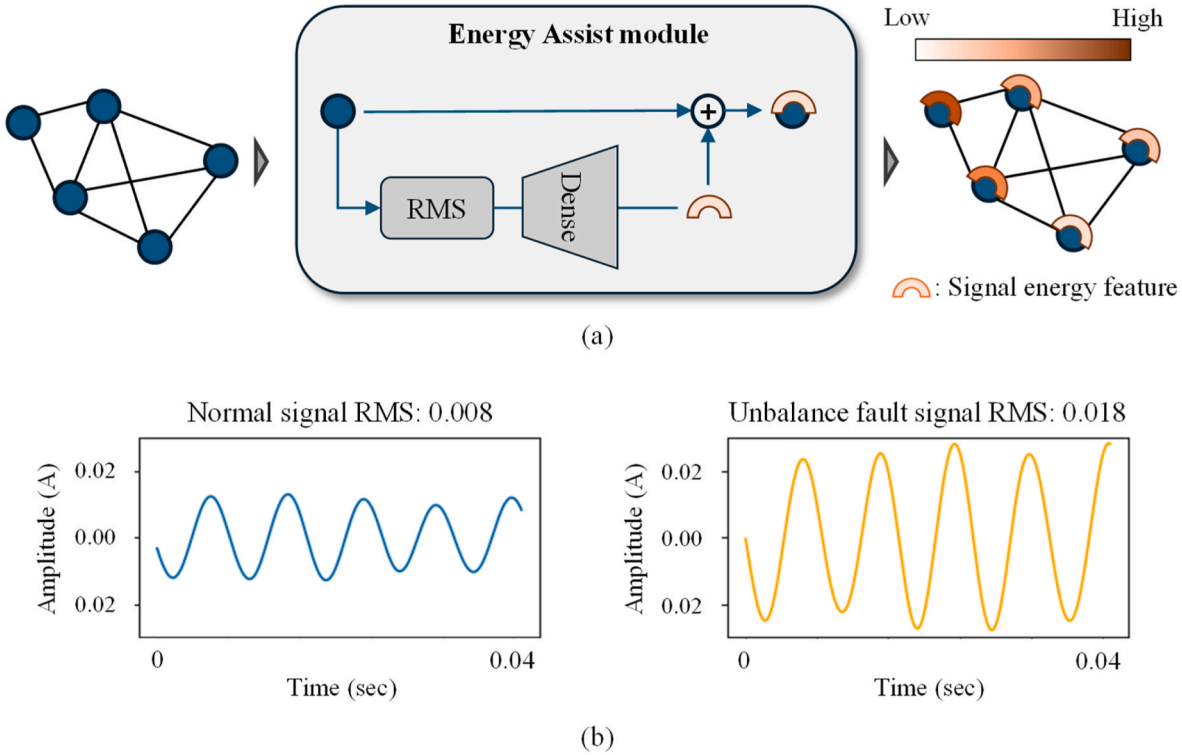


Fig. 7. Illustrations of the EA module:
(a) module architecture (b) fault-frequency-band signals comparison.

3.5. Model architecture and optimization

The detailed architecture and hyperparameter specifications of the proposed FBG-SEANet are summarized in Table 1. As illustrated in Fig. 3, the framework begins by enhancing the constructed FBG using the EA module, which embeds energy-shift features (RMS) into the nodes. The enriched graph is then processed through a hierarchical backbone consisting of two sequential blocks, each comprising a GCN layer and a pooling layer. To capture both local and global fault characteristics, the outputs from each block are individually processed by SR layers and then fused via element-wise summation. Finally, the fused representation is classified through a dense layer. In this study, three parameters are optimized for the proposed method: 1) hidden channels of the GCN layer, 2) pooling ratio for training the graphs, 3) number of attention heads in SR layer. For the first two parameters related to training, the hidden channels of the GCN layer and the pooling ratio in the pooling layer, were set to 1024 and 0.5, respectively. The hidden channels and pooling ratio were optimized using a grid search, with the hidden channel values set to 256, 512, 1024, and 2048, and the pooling ratio values set to 0.7, 0.6, 0.5, 0.4, and 0.3, respectively. For the number of heads in the SR layer, the performance was evaluated by varying the value from 2 to 6 in increments of 1. The value was set to 4, as no significant performance improvement was observed beyond this point. During the training process, the maximum number of epochs was set to 100, with early stopping applied after 10 epochs and a batch size of 64. The GCN layers parameters were updated using a stochastic gradient descent (SGD) optimizer. The initial learning rate was set as 0.01 with a momentum of 0.9.

Table 1
Model architecture specifications and hyperparameter settings.

Hyperparameter	Search space	Optimized value
1. Network Architecture		
GCN layer – hidden channels	{256, 512, 1024, 2048}	1024
Pooling layer – pooling ratio	{0.3, 0.4, 0.5, 0.6, 0.7}	0.5
SR layer – number of heads	{2,3,4,5,6}	4
2. Training Settings		
Optimizer	-	SGD
Initial learning rate	-	0.01
Momentum	-	0.9
Batch size	-	64
Max epochs	-	100 (Early stopping: 10)

4. Validation

The proposed FBG-SEANet was validated using three datasets: the KAIST rotating machinery dataset, the servo motor experimental dataset, and the UAM system experimental dataset. Each dataset was used to compare the performance of the proposed method against graph-based and state-of-the-art (SOTA) models. Additionally, analyses were conducted focusing on the unique characteristics of the dataset, such as fault modes and signal types. This enables a comprehensive evaluation of the algorithm's performance and interpretability. Table 2 shows the characteristics and corresponding analysis of each dataset. The experiments were conducted on a desktop system running Windows 11, equipped with a 12th Gen Intel Core i7-12700KF CPU and an NVIDIA GeForce RTX 3090 GPU. All implementations were developed using Python 3.8 and the PyTorch library. The detailed experimental settings and computational details for each case study are summarized in Table 3. The diagnosis accuracies of the proposed FBG-SEANet model and comparative models were calculated by averaging the results over five independent runs. All reported results for the test data are presented as the mean and standard error.

4.1. Case study 1: rotating machinery dataset

In this study, the diagnosis performance between the proposed model and conventional models was compared. Discussions on the results of graph similarity analysis, an ablation study, and feature visualization were followed to verify the effectiveness of the proposed

Table 2
Dataset characteristics and corresponding analyses for each case study.

Case study	Dataset characteristics	Validation objective	Analysis
1	Includes fault modes with relatively fewer distinctive fault characteristics as well as fault modes with clear unique fault frequencies	FBG-SEANet	(4.1.2) Accuracy comparison (4.1.4) Ablation study
		FBG	(4.1.3) Graph similarity analysis (4.1.4) Graph construction methods comparison
		EA module	(4.1.5) Feature visualization
2	Includes both electrical and mechanical fault modes	FBG-SEANet	(4.2.2) Accuracy comparison
		SR layer	(4.2.3) Attention score analysis
		FBG-SEANet	(4.2.4) Filter parameter analysis
3	Includes more diverse modal signals acquired from a more complex system	FBG-SEANet	(4.3.2) Accuracy comparison (4.3.3) Computational comparison

method. In particular, the discussion of this case study includes validation on the FBG and the EA module. This is because the dataset includes fault modes with relatively fewer distinctive fault characteristics, such as unbalance faults and misalignment faults, as well as fault modes with clear unique fault frequencies, such as bearing faults.

4.1.1. Experimental setup and FBG parameter settings

The KAIST rotating machinery testbed is open-source dataset provided by the Korea Advanced Institute of Science and Technology (KAIST) (Jung et al., 2023). Fig. 8 shows the overview of the KAIST rotating machinery testbed. The testbed operates at a speed of 3010 rpm, mainly consists of one brake, two bearing housings, rotors with a mass wheel, gearbox, torque meter, and a motor. The dataset contains four different fault modes: inner race fault (BPFI), outer race fault (BPFO) in bearing housing A, rotor unbalance faults and shaft misalignment faults. Vibration and current signals are collected under three different load conditions. Specifically, vibration data were measured in the x and y-direction at two bearing housings. Vibration and current signals are collected for all load conditions with a sampling rate of 25.6 kHz. The preprocessed sensor signals were segmented using a sliding window with a window size of 512 for graph construction. For each fault mode and the normal condition, 7000 samples were collected, resulting in a total of 35,000 samples. The dataset was divided into training, validation, and test sets in a 50:25:25 ratio. Table 4 summarizes the parameter settings used for the FBG construction. The parameters were determined based on the operating speed ($f_r \approx 50.1$ Hz). The bandwidth (B_{width}) was set to 30 Hz, which is the largest multiple of 10 satisfying the safety

Table 3
Experimental settings and computational details for each case study.

	Case study 1	Case study 2	Case study 3
Experiment hardware setting	12th Gen Intel Core i7-12700KF CPU NVIDIA GeForce RTX 3090 GPU Python 3.8 and the PyTorch library		
Number of sensors	5	4	7
Number of nodes per graph	170	136	182
Training samples	17500	5080	5000
Training time (min)	180	40	60
Inference time per sample (ms)	3	2	3

Table 4
Parameter settings for the FBG construction.

Parameter	Equation	Value
Rotational speed	-	3010 rpm
Rotating frequency (f_r)	-	50.1 Hz
Safety coefficient (α)	-	0.7
Frequency threshold ($F_{threshold}$)	-	1000
Bandwidth (B_{width})	$B_{width} \leq \alpha \cdot f_r$	30
Number of frequency bands (N)	$N = F_{threshold}/B_{width} + 1$	34

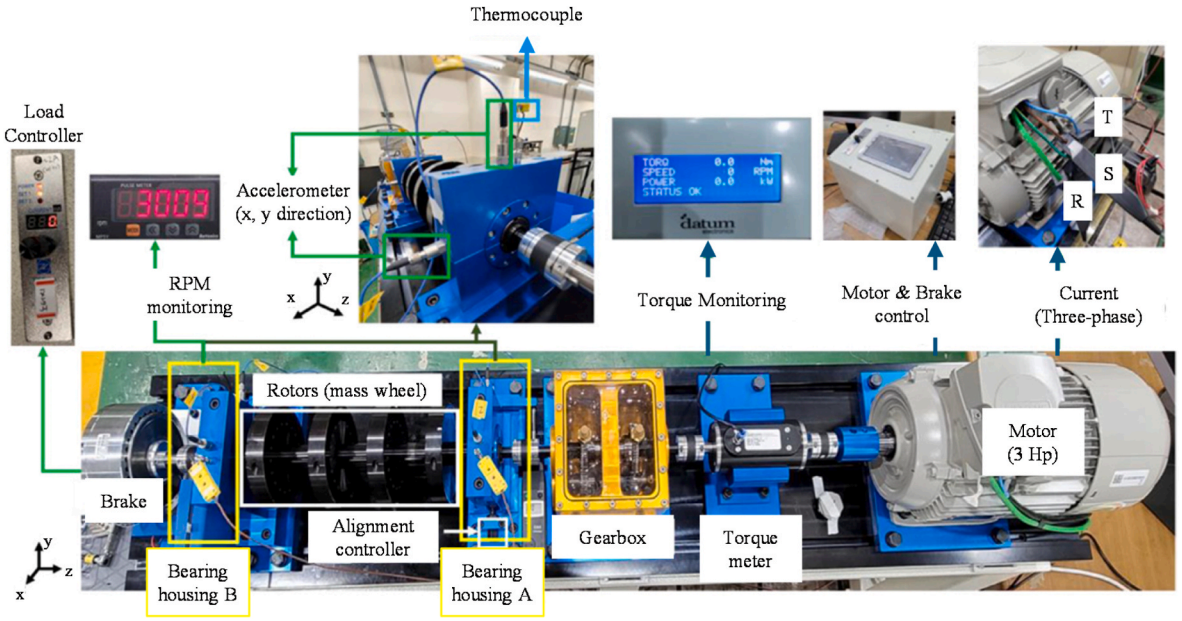


Fig. 8. System overview of the KAIST rotating machinery testbed.

Table 5
Accuracy comparison results.

Type	Model	Accuracy (%)
Conventional	GCN	79.18 ± 1.04
	GAT	82.96 ± 0.36
	GIN	89.22 ± 0.18
	GraphSAGE	92.70 ± 0.35
SOTA	MMHGAT (Zhang et al., 2023)	95.68 ± 0.47
	A-TSGNN (Wang et al., 2023)	79.87 ± 0.73
	TSFG (Wang et al., 2024a)	92.20 ± 0.64
	HTSFG (Sun et al., 2026)	95.58 ± 0.41
Proposed	FBG-SEANet	96.64 ± 0.49

margin condition relative to the fundamental frequency ($3010/60 \times 0.7 \approx 35.1$ Hz). Consequently, the number of frequency bands (N) was set to 34, calculated by taking the integer part of 1000/30 and adding 1, to sufficiently cover the frequency range up to 1000 Hz.

4.1.2. Comparison study

Table 5 shows the diagnostic accuracy comparison between the FBG-SEANet and other graph-level approaches. The comparison includes conventional methods commonly used in graph-based model training GCN, GAT, Graph Isomorphic Networks (GIN), and GraphSAGE combined with the KNNGraph graph construction method. In addition, SOTA graph-level models such as MMHGAT (Zhang et al., 2023), A-TSGNN (Wang et al., 2023), TFSG (Wang et al., 2024a), and HTSFG (Sun et al., 2026) were employed. Upon examining the results, the FBG-SEANet shows the highest accuracy, surpassing both existing conventional methods and SOTA models. Notably, conventional methods exhibited lower performance with accuracies ranging from 78% to 80%. In contrast, SOTA models, including GraphSAGE, demonstrated performance exceeding 94%. However, the proposed method outperformed all with an accuracy of 96.64%, showing higher performance. These results indicate that the graph construction method of the proposed approach accurately reflects the system's fault characteristics and that the model effectively learns these characteristics, thereby demonstrating the effectiveness of the proposed method.

4.1.3. FBG analysis through graph similarity score

This section evaluates the effectiveness of the proposed FBG in representing the system's fault characteristics for graph-level fault diagnosis. To analyze the FBG effectiveness in capturing the system's fault characteristics, a similarity score is used. This score measures the similarity between any two graphs by counting their number of common edges. A 'common edge' is defined as an edge that connects an identical pair of nodes present in both graphs being compared. The total count of these edges is then used as the score value.

$$\textit{Similarity score} = \mathrm{COUNT}(G_1, G_2) \tag{15}$$

Here, COUNT denotes the operation that count the number of common edges between two graphs G_1 and G_2. When a fault occurs in the system, the graph changes compared to normal situation. These changes can be reflected in the similarity score. The extent of the change depends on how prominently the characteristics of a specific fault are represented. Fault modes with significant characteristics result in lower similarity scores compared to those with less characteristics. Fig. 9 shows the similarity score result between the graphs for four fault modes (inner race fault, outer race fault, misalignment, unbalance) and the normal graph in the dataset. To measure the similarity score, 300 graphs were randomly extracted for the normal state and each fault mode from the training dataset. The similarity scores between the 300 graphs of each fault mode and the 300 normal graphs were calculated and averaged. The results showed that the bearing inner race fault had the lowest similarity score at 50.81, followed by the outer race fault with a similarity score of 52.77, which is higher than the inner race fault. Misalignment faults and unbalance faults, which are not bearing faults, exhibited significantly higher similarity scores of 57.41 and 58.83, respectively.

The following analysis results explain why different fault modes have different levels of similarity scores. Bearing faults have distinct fault frequencies, leading to a significant decrease in similarity scores. In contrast, misalignment and unbalance faults lack unique fault frequencies, resulting in minor graph changes and a slight decrease in similarity scores compared to the normal state. Furthermore, even

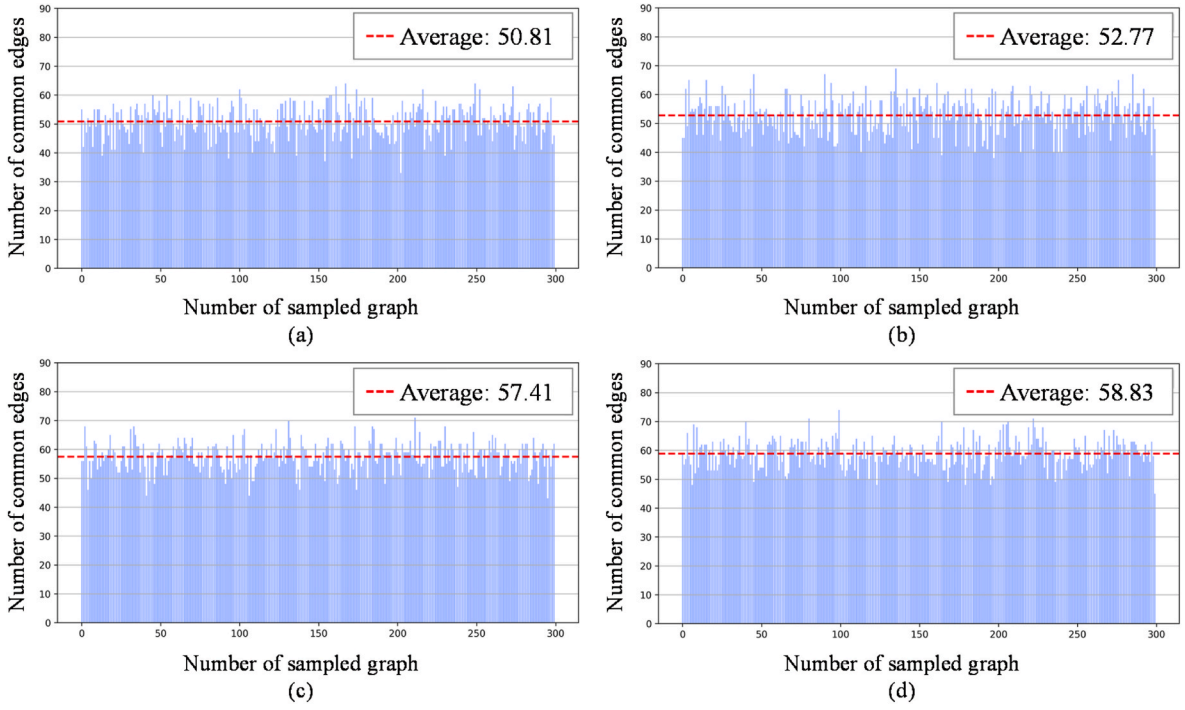


Fig. 9. Similarity score comparison for each fault mode:
(a) Inner race fault, (b) Outer race fault, (c) Misalignment fault, (d) Unbalance fault.

within the same category of bearing faults, the inner race fault has a lower similarity score than the outer race fault. This is because, unlike the outer race fault, the inner race fault not only has unique fault frequencies but also includes modulation signals, resulting in more distinctive graphs than those of the outer race fault (Smith and Randall, 2015). As the graph similarity analysis results align well with the actual fault characteristics, it can be concluded that the proposed FBG effectively reflects the system's fault.

4.1.4. Graph construction methods comparison

In this section, the proposed FBG is compared with other commonly used graph construction methods for system-level fault diagnosis, including KNNGraph, RadiusGraph, and PathGraph (Li et al., 2022a). These conventional methods typically use raw sensor signals directly for node embedding. Table 6 shows the diagnostic performance of these conventional methods and the proposed FBG, all trained using the same GCN model to ensure a fair comparison. The results show that the model trained with the proposed FBG achieves diagnosis accuracy exceeding 90%, whereas the conventional graph construction methods reach only around 80%. This performance gap arises because directly using raw sensor signals for node construction fails to effectively encode fault-specific characteristics into the graph. In particular, operational components tend to dominate the raw sensor signals, making it difficult for subtle fault features to stand out. Moreover, some fault characteristics are distributed across multiple frequency bands, but in raw signals these components are often overlapped. As a result, such frequency-domain information cannot be effectively captured in the graph when using raw signals alone. In contrast, the proposed FBG effectively captures these subtle and frequency-distributed fault features in the graph, which leads to its improved diagnostic performance.

4.1.5. Ablation study and feature visualization

An ablation study was conducted to demonstrate the contribution of each component of the proposed method; the results are shown in Table 7. The results confirm that the proposed FBG, EA module and SR layer contribute to improved diagnosis accuracy. These results should be interpreted not only as numerical performance comparisons, but also as functional validation of the framework design. Replacing the KNN graph with the proposed FBG increases the diagnosis accuracy from 90.0% in Model 1 to 92.1% in Model 2, demonstrating the benefit of preserving fault-related frequency structure in the graph representation. The SR layer further improves the accuracy from 92.1% in Model 2 to 93.7% in Model 3 without the EA module, and from 95.3% in Model 4 to 96.6% in Model 5 with the EA module. These consistent gains indicate that the SR layer contributes to diagnosis performance by emphasizing fault-relevant node information during graph-level readout. This role is important because uniform aggregation can dilute diagnostically informative sensor-band components with less relevant nodes. Finally, the comparison between Model 3 and Model 5 shows that the EA module improves the accuracy from 93.7% to 96.6%, confirming its effectiveness for faults characterized by energy shifts.

To further analyze the effectiveness of the EA module, t-distributed stochastic neighbor embedding (t-SNE) map visualization was also performed. The feature visualization for Model 3 and Model 5 are shown in Fig. 10(a) and (b), respectively. The Model 3 t-SNE visualization in Fig. 10(a), shows that the fault modes are generally well classified. However, differences between misalignment, unbalance, and normal are not evident. This is because misalignment and unbalance do not have unique fault frequencies and are characterized by changes in the energy of rotational frequencies, making their fault characteristics relatively weak. As a result, in the case of bearing faults with clear unique fault frequencies, the features are distinctly separated, whereas the features of misalignment and unbalance faults are closely distributed with normal features. In contrast, the t-SNE map for Model 5, depicted in Fig. 10(b), shows clear differences between misalignment, unbalance, and normal. This indicates that the EA module, which adds energy features of signals, helps clearly distinguish misalignment and unbalance faults from normal faults. In other words, it demonstrates that the EA module is necessary to clearly differentiate even those faults that do not have unique fault frequencies but are characterized by changes in the energy of pre-existing rotational frequencies. Furthermore, the training times of Model 3 and Model 5 showed no notable difference, indicating that the EA module introduces negligible computational overhead while effectively improving the diagnosis of energy-shift-type faults. This is because the EA module adds only lightweight node-wise operations, namely the computation of a scalar root-mean-square value from the time-domain signal and its projection through a single dense layer. In contrast, the dominant computational cost of the overall framework arises from the graph convolution layers and the multi-head attention mechanism in the SR layer. Therefore, the EA module provides a favorable practical trade-off by improving fault separability without meaningfully increasing the computational cost.

Table 6
Comparison results of graph construction methods.

Graph type	Train model	Accuracy (%)
KNNGraph	GCN	79.18 ± 1.04
RadiusGraph		80.9 ± 7.34
PathGraph		80.6 ± 6.59
FBG (Proposed)		92.2 ± 2.22

Table 7
Ablation study results.

Model type	Graph type	w/o SR layer	w/o EA module	Accuracy (%)
Model 1	KNNGraph (Li et al., 2022a)	X	X	90
Model 2	Frequency-band	X	X	92.1
Model 3	Frequency-band	O	X	93.7
Model 4	Frequency-band	X	O	95.3
Model 5 (Proposed)	Frequency-band	O	O	96.6

4.2. Case study 2: motor dataset

In this study, diagnosis accuracies of the proposed model and comparison models were evaluated. Next, the results of attention score analysis were discussed to verify the effectiveness of the SR layer. The reason is that this dataset includes both electrical and mechanical fault modes. It is known that vibration signals are more sensitive for detecting mechanical faults, while current signals are more sensitive for detecting electrical faults (Gangsar and Tiwari, 2020). Therefore, in this section, using the motor dataset, the role of the SR layer was investigated through self-attention scores, and the signals with high attention scores for each fault mode were identified. The multi-head fusion strategy adopted in the SR layer was also examined by comparing three different fusion approaches. Additionally, to assess the robustness of the proposed method to its signal processing settings, a sensitivity analysis was conducted on the notch filter and bandwidth selection used in the FBG construction.

4.2.1. Experimental setup and FBG parameter settings

The motor system depicted in Fig. 11 is a surface-mounted permanent magnet synchronous motor (PMSM) with a rated power of 200 W, a torque capacity of 5.5 Nm, and 20 poles. Position control was implemented using an incremental encoder offering a resolution of 4000 pulses per revolution. Stator current signals from the three phases were acquired via Tektronix A622 current probes at a sampling rate of 128,000/s. To evaluate fault diagnosis performance, three fault types were artificially introduced: a stator inter-turn short (SIS) as an electrical fault, and misalignment and bearing faults as representative

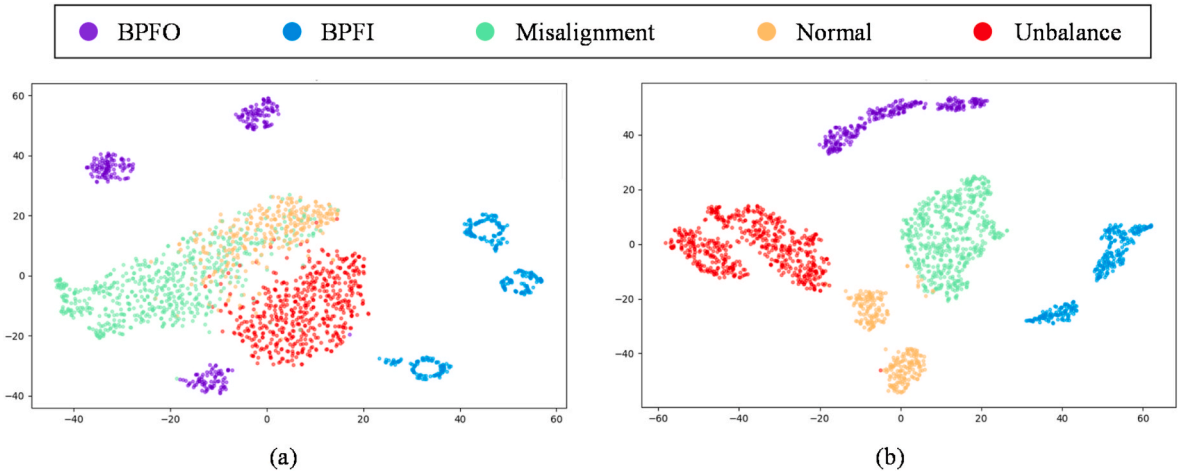


Fig. 10. t-SNE visualization for two different models: (a) Model 3, (b) Model 5.

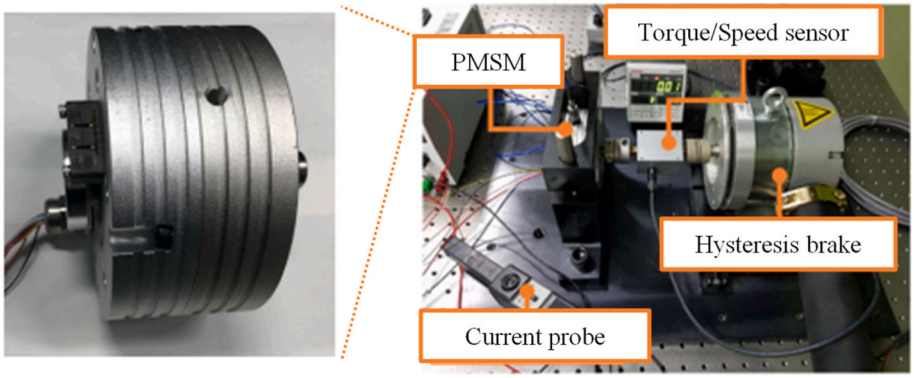


Fig. 11. System overview of the SNU motor testbed.

mechanical faults. The SIS fault was generated by winding exposed stator coils together, whereas the misalignment condition was emulated by vertically offsetting the shaft by 2 mm. For the operating conditions, the motor was rotated at 3000 rpm with a torque setting of 5.5 Nm. The preprocessed sensor signals were segmented using a sliding window with a length of 512. As a result, 2540 samples were generated for each fault mode and normal condition, yielding a total of 10,160 samples. These data were split into training, validation, and test sets in a ratio of 50:25:25.

Table 8 summarizes the parameter settings used for the FBG construction. The parameters were determined based on the operating speed ($f_r \approx 50$ Hz). The bandwidth (B_{width}) was set to 30 Hz, which is the largest multiple of 10 satisfying the safety margin condition relative to the fundamental frequency ($3000/60 \times 0.7 \approx 35$ Hz). Consequently, the number of frequency bands (N) was set to 34, calculated by taking the integer part of 1000/30 and adding 1, to sufficiently cover the frequency range up to 1000 Hz.

Table 8
Parameter settings for the FBG construction.

Parameter	Equation	Value
Rotational speed	-	3000 rpm
Rotating frequency (f_r)	-	50 Hz
Safety coefficient (α)	-	0.7
Frequency threshold ($F_{threshold}$)	-	1000
Bandwidth (B_{width})	$B_{width} \leq \alpha \cdot f_r$	30
Number of frequency bands (N)	$N = F_{threshold}/B_{width} + 1$	34

Table 9
Accuracy comparison results.

Type	Model	Accuracy (%)
Conventional	GCN	82.70 ± 4.08
	GAT	83.16 ± 6.04
	GIN	77.84 ± 12.49
	GraphSAGE	95.71 ± 0.36
SOTA	MMHGAT (Zhang et al., 2023)	95.24 ± 1.37
	A-TSGNN (Wang et al., 2023)	94.07 ± 1.68
	TSFG (Wang et al., 2024a)	94.94 ± 1.86
	HTSFG (Sun et al., 2026)	94.31 ± 5.31
Proposed	FBG-SEANet	96.43 ± 0.31

4.2.2. Comparison study

Table 9 shows the comparison results. For a fair comparison, the same conventional and SOTA models used in case study 1 were employed. Upon examining the results, the FBG-SEANet achieved the highest accuracy, surpassing both existing conventional methods and SOTA models. Notably, conventional models showed lower performance with accuracies around 80%, while SOTA models, including GraphSAGE, demonstrated performance exceeding 94%. The proposed method outperformed all comparison models, achieving an accuracy of 96.43%. These results indicate that the proposed graph construction method reflects the system's fault characteristics and that the model effectively learns these characteristics, thereby demonstrating the effectiveness of the proposed method.

To further analyze the diagnosis results, the confusion matrix and t-SNE visualization of the proposed FBG-SEANet are presented in Fig. 12.

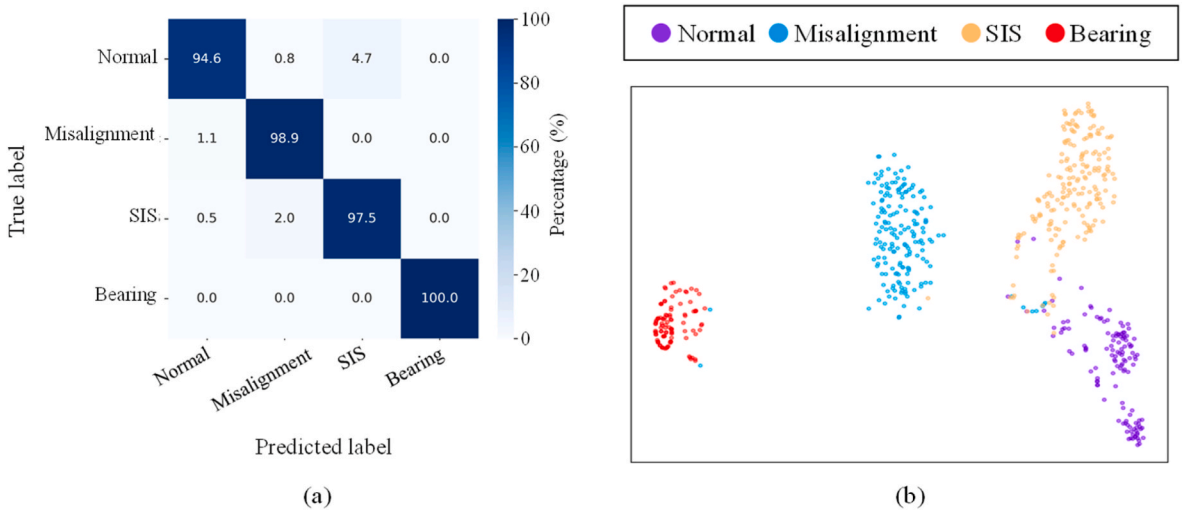


Fig. 12. Visualization of fault diagnosis results for the motor system dataset: (a) confusion matrix, (b) t-SNE visualization.

Table 10
Results of the self-attention score analysis.

	1	2	3	4
SIS (Electric fault)	Current_V	Current_U	Current_W	Vibration_X
Misalignment (Mechanical fault)	Vibration_X	Current_U	Current_W	Current_V
Bearing (Mechanical fault)	Vibration_X	Current_V	Current_W	Current_U

As shown in Fig. 12(a), the proposed method achieved high classification performance across all fault modes, with the bearing fault being classified without misclassification. The few misclassifications were mainly observed among the normal, misalignment, and SIS classes, indicating that these classes contain partially overlapping signal characteristics. Fig. 12(b) shows that the learned graph-level features form distinguishable clusters for each fault mode. In particular, the bearing fault cluster is clearly separated from the other classes, which is consistent with the confusion matrix result. These visualization results provide a more detailed interpretation of the diagnosis results by showing both class-wise diagnostic performance and feature-level separability.

4.2.3. SR layer analysis

The proposed method includes an SR layer to effectively assign importance to different signals according to the fault mode. To analyze its effectiveness, the self-attention scores of all nodes were examined for each fault mode, where a higher score indicates a greater contribution to the diagnosis. Table 10 ranks the signal types by their diagnosis importance, determined by the frequency of their corresponding nodes appearing among the top 20 nodes with the highest attention scores. The signal ranked first is the one most frequently identified as critical for diagnosing a specific fault. The results show that the SIS fault, an electrical fault, exhibits high attention scores for current signals while mechanical faults such as misalignment and bearing faults show the highest attention scores for a vibration signal. These results indicate that the SR layer effectively reflects the importance of signals according to fault modes in graph representation. This role is particularly important in graph-level fault diagnosis, because the readout stage compresses node-level information into a single graph embedding. If all nodes are aggregated uniformly, diagnostically important information from highly sensitive sensor-band components can be diluted by less informative nodes. By assigning fault-dependent importance scores before aggregation, the SR layer enables the final graph representation to preserve the most relevant fault characteristics while suppressing redundant contributions. Therefore, the contribution of the SR layer extends beyond a simple weighting operation; it functions as a fault-sensitive graph-level readout mechanism and an interpretable node-importance estimator. This leads to a more discriminative and physically interpretable graph representation for fault diagnosis.

The interpretability of the attention scores is closely related to the multi-head fusion strategy used in the SR layer. To examine this, three fusion strategies were compared: (1) averaging (proposed), (2) concatenation followed by linear projection, and (3) learnable weighted fusion. A comparison of the multi-head fusion strategies used in the SR layer is presented in Fig. 13. The corresponding accuracies are 96.43%, 94.37%, and 96.61%, respectively. Importantly, the role of the SR layer differs from that of standard multi-head attention in transformers. In standard multi-head attention, concatenation and projection are used to fuse head-wise feature representations. In contrast, the SR layer aggregates head-wise attention weights into a single scalar importance score for each node. Under this setting, averaging provides a direct and parameter-free way to aggregate head-wise evidence into a unified node-importance score.

The comparison results show that averaging substantially outperforms concatenation-projection, suggesting that introducing an additional feature-space transformation is less suitable for this scalar weight aggregation step. Such a design may also increase the risk of overfitting, particularly in relatively small-graph settings, by adding unnecessary capacity to a module whose role is only to produce a scalar importance score. Learnable weighted fusion achieves a slightly higher accuracy (+0.18%), but this gain is marginal relative to the additional learnable parameters and computational cost introduced during training. Therefore, averaging was adopted as the default fusion strategy because it offers a favorable trade-off among accuracy, simplicity, efficiency, and interpretability.

4.2.4. Sensitivity analysis of filter parameters

The proposed FBG construction relies on two signal-processing stages: the OCF with a notch filter and band-pass filtering with a specified bandwidth. To examine the sensitivity of the proposed method to these filter parameters, two analyses were conducted. Table 11

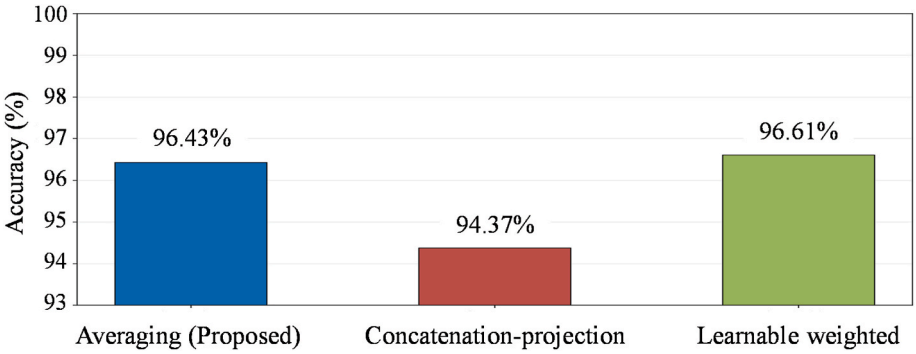


Fig. 13. Comparison of multi-head fusion strategies in the SR layer.

Table 11
Sensitivity analysis results for filter parameters.

Analysis	Type/Bandwidth (B_{width})	Accuracy (%)
OCF ablation	With OCF (proposed)	96.43 ± 0.31
	Without OCF	82.85 ± 4.27
Bandwidth sensitivity analysis	10 Hz	87.07 ± 0.90
	20 Hz	90.67 ± 0.67
	30 Hz (proposed)	96.43 ± 0.31
	40 Hz	94.27 ± 0.30
	50 Hz	95.07 ± 0.42

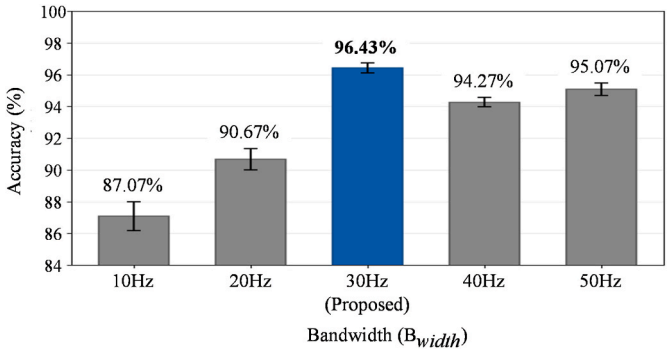


Fig. 14. Comparison of diagnosis accuracy across different bandwidth settings.

summarizes the sensitivity analysis results for the filter parameters, and Fig. 14 illustrates the comparison of diagnosis accuracy across different bandwidth settings.

First, the effect of the notch filter was evaluated by comparing the proposed method with and without the OCF. As shown in Table 11, removing the OCF reduces the accuracy from 96.43% to 82.85%. This can be attributed to the dominant operational components in the raw signals, which mask subtle fault-related features across the frequency bands and make it more difficult for the resulting FBG to preserve diagnostically useful fault characteristics. The center frequency of the notch filter is determined by the known operating frequency of the target system, rather than being arbitrarily selected. Therefore, the OCF improves fault sensitivity while limiting the risk of suppressing fault-relevant frequency components.

Second, the sensitivity to bandwidth selection was examined based on the bandwidth selection rule introduced in Section 3.2.2. For the system considered in this case study, 30 Hz was selected as the bandwidth, and additional experiments were conducted by varying the bandwidth from 10 Hz to 50 Hz in increments of 10 Hz to evaluate the robustness. As shown in Table 11 and Fig. 14, the selected bandwidth of 30 Hz achieves the highest accuracy.

When the bandwidth is smaller than the selected setting, such as 10 Hz and 20 Hz, two issues arise. First, narrower band-pass filters can lead to longer impulse responses relative to the analysis window, which may introduce spectral leakage and transition-band distortion. This can reduce the fidelity of the filtered signal and lead to information loss in the node features. Second, smaller bandwidths produce a significantly larger number of graph nodes, with 404 nodes at 10 Hz compared with 136 nodes at 30 Hz. This increases model complexity and potentially increases the risk of overfitting.

Conversely, when the bandwidth exceeds the selected range, such as 40 Hz and 50 Hz, each frequency band spans a wider spectral range, causing multiple harmonic components to be merged into a single node. As a result, fault-characteristic energy becomes diluted across a broader band, reducing the discriminative power of each node feature. This weakens the model's ability to capture inter-harmonic dependencies that are important for distinguishing different fault modes. Overall, the results indicate that the bandwidth selection rule is appropriate for preserving diagnostically relevant spectral structure while maintaining a compact graph representation.

4.3. Case study 3: UAM system dataset

In this study, a comparison of accuracy between the proposed model and comparison models was conducted. The purpose of this case study was to verify the effectiveness of the proposed method by utilizing more diverse multi-modal signals acquired from a more complex system compared to the previous case studies.

4.3.1. Experimental setup and FBG parameter settings

The UAM testbed shown in Fig. 15 includes three BLDC motors with blades, a tilting linkage with an actuator, and a wing. Specifically, the tilting mechanism used in the lift and cruise mode is implemented through a linkage with an actuator, allowing the front tilt motor to tilt from 0° to 90°. The side motor at the wingtip and the back motor are not tiltable that are used for assistance during the ascent. The pylon composed of tilt motor, back motor, and tilting mechanism is connected to the lower part of the wing via bolt-type load cells, enabling the measurement of the load acting between the pylon and the wing. The T-motor AT 7224 KV190 with a maximum thrust of 17.7 kg is used as the BLDC motor and the wings are made of woven CFRP(carbon fiber/epoxy) composite material and have two ribs attached internally.

The testbed is equipped with current and vibration sensors, strain gauges, and load sensors, allowing for the acquisition of various multi-modal signals. DEWESOFT-R8 IOLITE DAQ was used to collect these multi-modal signals, with a sampling rate set to 10,000 Hz. The acquired data included one channel of current signals of the tilt motor, one channel of single axis vibration signal generated by the tilt motor, and strain changes in the wing during operation obtained through two strain gauges. Additionally, two bolt-type load sensors connecting the pylon and the wing were used to capture load changes between the pylon and the wing during operation.

Experiments simulating the operation of the UAM were conducted,

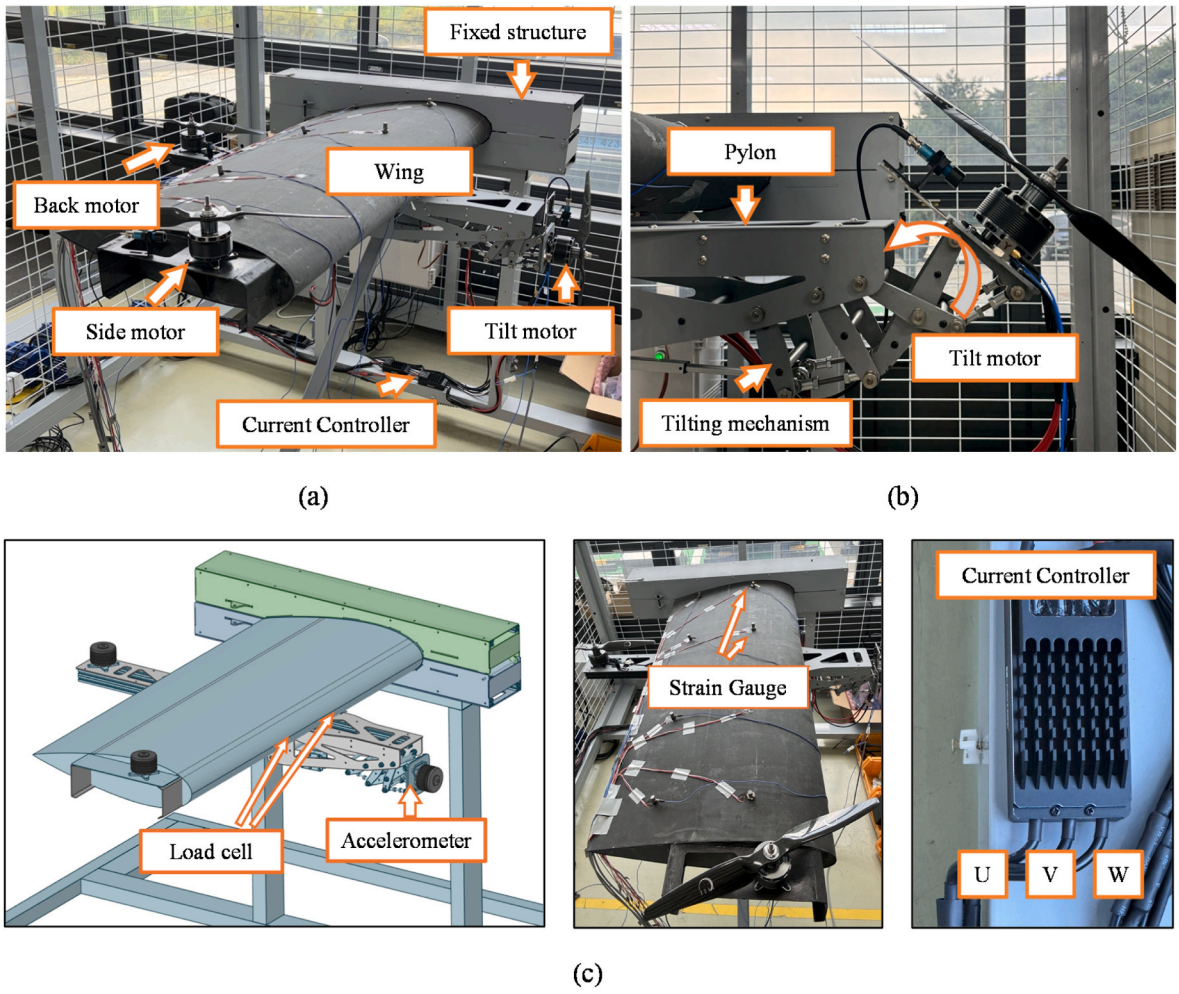


Fig. 15. System overview of the SNU UAM testbed:
(a) testbed components (b) tilt motor (c) sensor signal acquisition system.

Table 12
Parameter settings for the FBG construction.

Parameter	Equation	Value
Rotational speed	-	4000 rpm
Rotating frequency (f_r)	-	66.7 Hz
Safety coefficient (α)	-	0.7
Frequency threshold ($F_{threshold}$)	-	1000
Bandwidth (B_{width})	$B_{width} \leq \alpha \cdot f_r$	40
Number of frequency bands (N)	$N = F_{threshold}/B_{width} + 1$	26

where the three motors rotated at a constant speed of 4000 rpm, and the tilt motor followed a triangular drive profile by tilting from 0° to 90° and then returning to 0°. Additionally, the experiments were performed with two different system fault modes, including inter-turn short fault in the tilt motor and demagnetization fault. The preprocessed sensor signals were segmented using a sliding window of length 512, resulting in 2500 samples for the normal condition and for each fault mode, yielding a total of 10,000 samples. These data were split into training, validation, and test sets in a ratio of 50:25:25.

Table 12 summarizes the parameter settings used for the FBG construction. The parameters were determined based on the operating speed ($f_r \approx 66.7$ Hz). The bandwidth (B_{width}) was set to 40 Hz, which is the largest multiple of 10 satisfying the safety margin condition relative to the fundamental frequency ($4000/60 \times 0.7 \approx 46.7$ Hz). Consequently, the number of frequency bands (N) was set to 26, calculated by taking the integer part of 1000/40 and adding 1, to sufficiently cover the frequency range up to 1000 Hz.

Table 13
Accuracy comparison results.

Type	Model	Accuracy (%)
Conventional	GCN	96.79 ± 0.87
	GAT	97.76 ± 0.62
	GIN	94.85 ± 2.66
	GraphSAGE	92.89 ± 3.54
SOTA	MMHGAT (Zhang et al., 2023)	96.15 ± 0.73
	A-TSGNN (Wang et al., 2023)	81.26 ± 2.97
	TSFG (Wang et al., 2024a)	96.90 ± 0.62
	HTSFG (Sun et al., 2026)	95.15 ± 0.74
Proposed	FBG-SEANet	98.68 ± 0.23

4.3.2. Comparison study

Table 13 presents the comparison results. For a fair comparison, the same classical models and state-of-the-art models used in Case Studies 1 and 2 were employed. Upon examining the results, the FBG-SEANet achieved the highest accuracy, surpassing both existing conventional methods and SOTA models. Specifically, the conventional classical models showed relatively high performance with accuracies above 95%, and except for the A-TSGNN model, SOTA models demonstrated performances exceeding 96%. However, the proposed method outperformed all with an accuracy of 98.68%. Notably, the proposed method had the lowest variation at 0.23 among the comparison models,

Table 14
Computational comparison with baseline models.

Model	Training time (min)	Inference time per sample (ms)	Accuracy (%)
GCN	40	3	96.79 ± 0.87
GAT	50	2	97.76 ± 0.62
GIN	45	2	94.85 ± 2.66
GraphSAGE	35	2	92.89 ± 3.54
FBG-SEANet	60	3	98.68 ± 0.23

indicating that the proposed method was trained stably and effectively learned the system's fault characteristics. Therefore, it was confirmed that the FBG-SEANet effectively learned the fault characteristics of the system by utilizing diverse modal signals acquired from a complex system, thereby demonstrating the proposed method's effectiveness in multi-sensor, multi-modal fusion.

4.3.3. Computational cost and scalability analysis

To further evaluate the practical feasibility of the proposed method, the computational cost was compared with conventional baseline graph models using the UAM system dataset. Case Study 3 was selected for this analysis because it contains the largest graph among the three case studies, with seven sensor channels. All models were evaluated under the same hardware and implementation settings, and the results are summarized in Table 14. The results show that FBG-SEANet requires a slightly longer training time than the baseline graph models. This increase is mainly attributed to the frequency-band graph representation and the sensitivity-aware readout layer, which introduce additional graph-level feature learning compared with conventional graph models. However, the inference time of FBG-SEANet remains 3 ms per sample, which is comparable to those of the baseline models. In addition, FBG-SEANet achieves the highest diagnosis accuracy of 98.68 ± 0.23%, indicating that the additional training cost provides improved diagnosis performance while maintaining practical inference efficiency.

Regarding scalability, the number of nodes in the proposed FBG is determined by M × N, where M is the number of sensor channels and N is the number of frequency bands. Therefore, under a fixed frequency-band setting, the graph size increases linearly with the number of sensors. It should be noted that this increase in computational cost is not unique to the proposed method; baseline graph models also require additional computation as the number of sensor nodes increases. However, the proposed method is designed to exploit the increased sensor information more effectively by constructing sensor-band-specific nodes and applying the SR layer to learn fault-dependent sensor-band importance. Therefore, although the training cost may increase as more sensors are added, the proposed framework has the potential to provide stronger sensor fusion capability than conventional graph models in large-scale multi-sensor systems. Overall, these results demonstrate that FBG-SEANet achieves the highest diagnosis accuracy while maintaining reasonable training time and millisecond-level inference, indicating a favorable trade-off between computational cost and diagnostic performance.

5. Conclusion

This study introduces the frequency band graph-based sensor fusion with sensitivity-aware energy assist network (FBG-SEANet), a novel graph-level sensor fusion method designed to enhance fault diagnosis in machinery systems. First, the proposed frequency-band graph (FBG) is constructed to effectively reflect the fault characteristics across multi-sensor signals in the input graph. Then, the proposed sensitivity-aware energy assist network (SEANet) is employed to effectively represent the constructed graph. Specifically, the energy assist (EA) module in SEANet enriches node embedding features with energy-based features helping diagnose fault modes characterized by energy changes in pre-existing frequency components. In addition, the sensitivity-aware readout (SR) layer is utilized to capture the relative importance among sensors, thereby preventing information dilution and reducing redundancy.

To verify the FBG-SEANet performance, three datasets involving an open rotating machinery dataset, a motor experimental dataset, and an UAM system experimental dataset were used for validation. In each study, the proposed method outperformed conventional graph-based models (i.e., GCN, GAT, GIN and GraphSAGE) and state-of-the-art (SOTA) methods (i.e., MMHGAT, A-TSGNN, TSFG, and HTSFG), achieving higher accuracy with lower variances. For further analysis of the FBG-SEANet, a similarity score was used to demonstrate the FBG superiority in representing system fault characteristics. An ablation study was conducted to verify the significant contributions of each component of the proposed method. Furthermore, feature visualization is demonstrated to highlight the improved differentiation enabled by the EA module. Overall, the proposed method effectively fuses multi-modal signal information for fault diagnosis, addressing the limitations of conventional methods.

Although the proposed method demonstrates strong capabilities in graph construction and representation, achieving high fault diagnosis performance, it remains limited to stationary operating conditions due to the reliance on fixed-frequency band-pass filtering for FBG construction, whose parameters are derived from a constant rotational speed. Under non-stationary operating conditions, the frequency-domain fault characteristics shift with the varying rotational frequency, making it difficult to construct an FBG that consistently represents the intended fault information using fixed band-pass filters. Therefore, future work will focus on extending FBG-SEANet to non-stationary operating conditions by redesigning the graph construction process through techniques such as order tracking or short-time Fourier transform (STFT)-based adaptive band construction, incorporating additional sensor modalities, and evaluating its applicability and robustness in diverse industrial environments.

CRediT authorship contribution statement

Sang Kyung Lee: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Hyeongmin Kim:** Validation, Methodology, Conceptualization. **Minseok Chae:** Writing – review & editing, Validation, Data curation. **Hansoo Kim:** Writing – review & editing, Methodology, Conceptualization. **Joonho Yang:** Visualization, Conceptualization. **Heonjun Yoon:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Byeng D. Youn:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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